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Improving Front-Back Depth Reasoning in Vision-Language Models via Parameter-Efficient Spatial Cue Fine Tuning

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1

Motivation

- Vision-language models perform well on general multimodal tasks but still struggle with **spatial reasoning in partially occluded scenes**.
- We focus on **front/back depth reasoning**: given two localized objects A and B in an image, determine whether A is in front of B or behind it.
- Our goal is to test whether lightweight structured spatial cues can improve this reasoning without full model retraining.

2

Related Works

- VLMs still struggle with **spatial reasoning** despite strong general VQA performance.
- Recent benchmarks expose failures in embodied, multi-view, dynamic, and perspective-taking spatial tasks.
- Our work follows explicit-structure methods, but focuses narrowly on **occlusion-based** front/back reasoning using **lightweight spatial cues**.

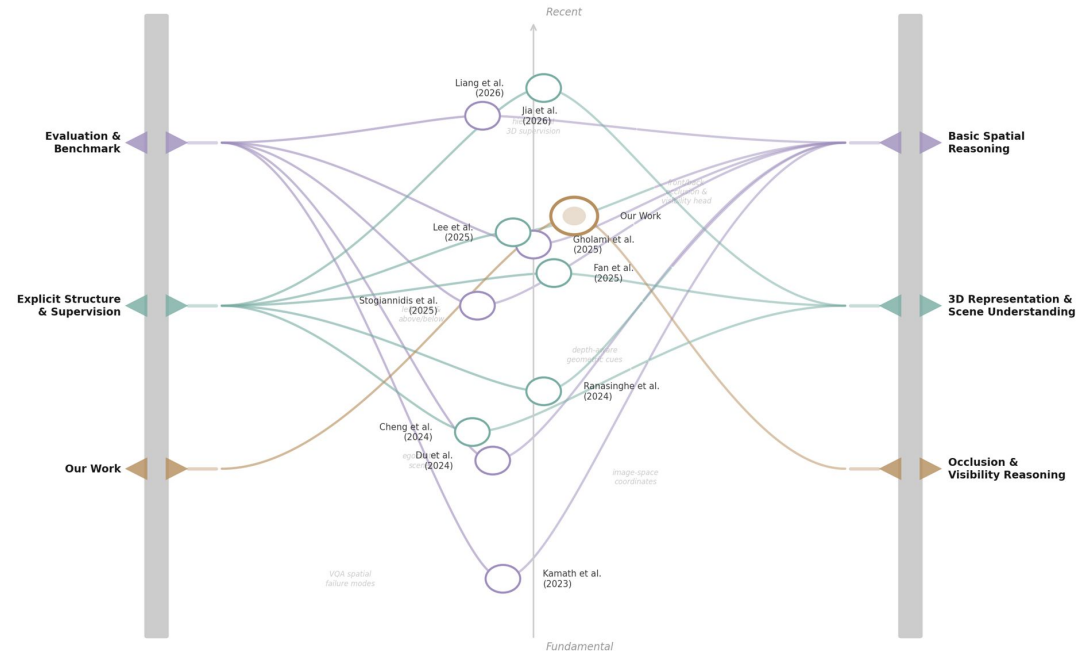


Figure 1, Territory Map of Related Works

Source Data: InstaOrder paired with COCO 2017 images.

[1] Lin, Tsung-Yi, et al. "Microsoft coco: Common objects in context." *European conference on computer vision*. Cham: Springer International Publishing, 2014. <https://cocodataset.org/>

[2] Lee, Hyunmin, and Jaesik Park. "Instance-wise occlusion and depth orders in natural scenes." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022. <https://github.com/SNU-VGILab/InstaOrder>

Split	# Samples	% front	% behind
Train	26,046	50.0	50.0
Val	2,746	50.0	50.0
Test	2,952	50.0	50.0

Table 1: Numbers of samples



Figure 2, Original Image [1]

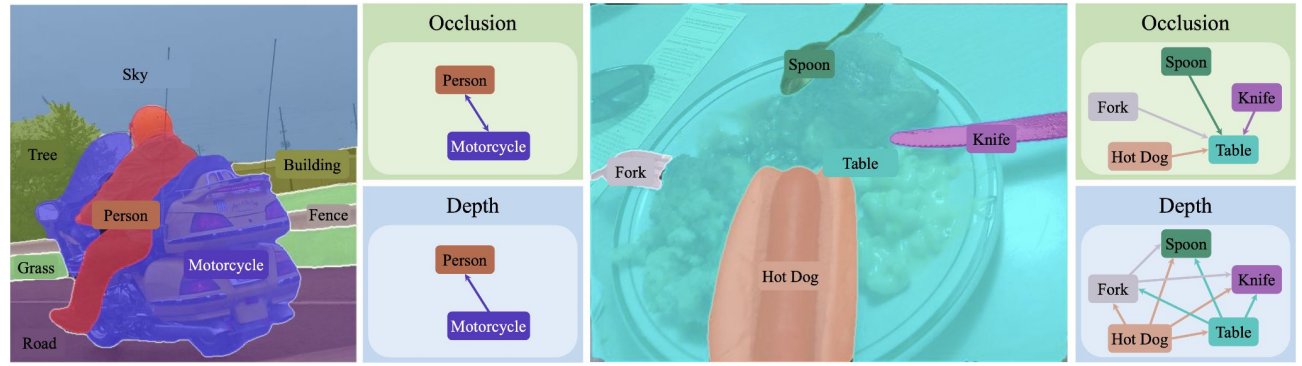


Figure 3, Occlusion Relationships from InstaOrder Dataset [2]

Our Dataset

- **Directionally balanced pairs:** each unordered pair is emitted twice with flipped labels.
- **Image-disjoint split:** all pairs from the same image stay in the same split.
- **Marked-image variant:** queried bounding boxes are rendered and labeled A and B.

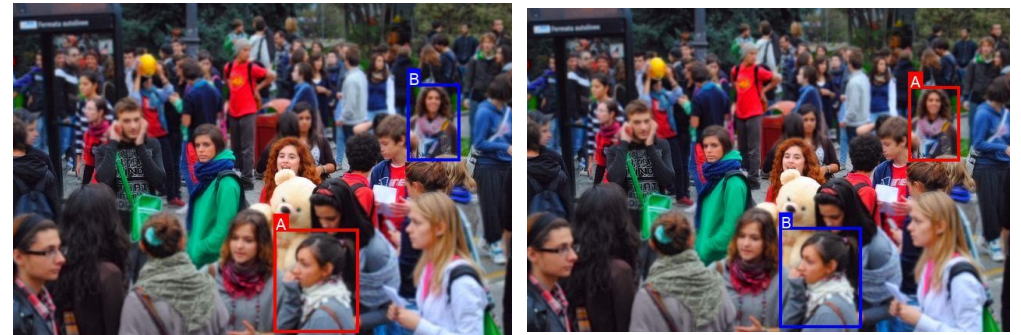


Figure 4, Processed Image

Metric	Description
Raw Accuracy	Fraction of test queries answered correctly on the balanced test set.
Accuracy by label	Accuracy computed separately on <i>front</i> and <i>behind</i> queries, which helps detect residual label bias.
Accuracy by overlap	Accuracy split by whether the two queried bounding boxes overlap in 2D, separating pure depth-ordering cases from stronger occlusion-style cases.
Consistency Score	Fraction of unordered pairs $\{a, b\}$ whose two directional predictions are mutually consistent, i.e., one is <i>front</i> and the other is <i>behind</i> .
Pair Accuracy	Fraction of unordered pairs $\{a, b\}$ for which both directional queries are answered correctly.
Front-Bias Rate	Fraction of test queries for which the model answers <i>front</i> , regardless of ground truth. This diagnoses prior-driven guessing.
Rationale Quality Score	Score assigned by a judge model (e.g., GPT-4o or Claude) to each generated rationale, based on whether it is grounded in the image, whether it uses valid depth cues, and whether it contains hallucination.

Table 2, Evaluation Metrics

Model: Qwen2.5-7B-Instruct; Qwen2.5-72B

Adaption: QLoRA, rank 64, α 128, dropout 0.05, lr 1×10^{-4}

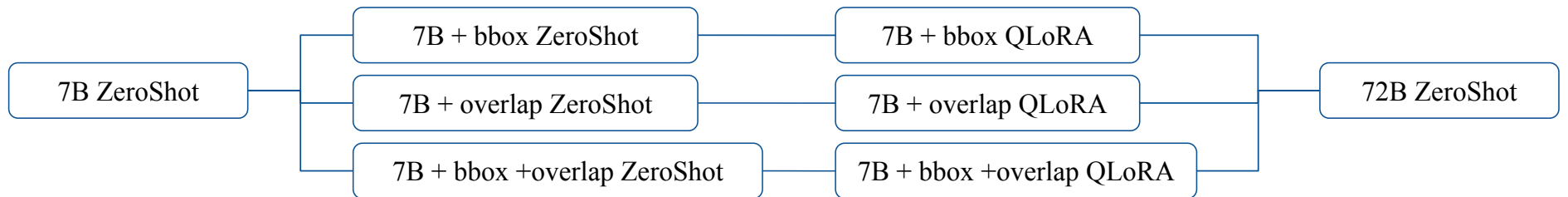


Figure 5, Experiment Settings

(I) Inputs

(II) Prompt Construction

(III) Training and Inferencing

(IV) Evaluation

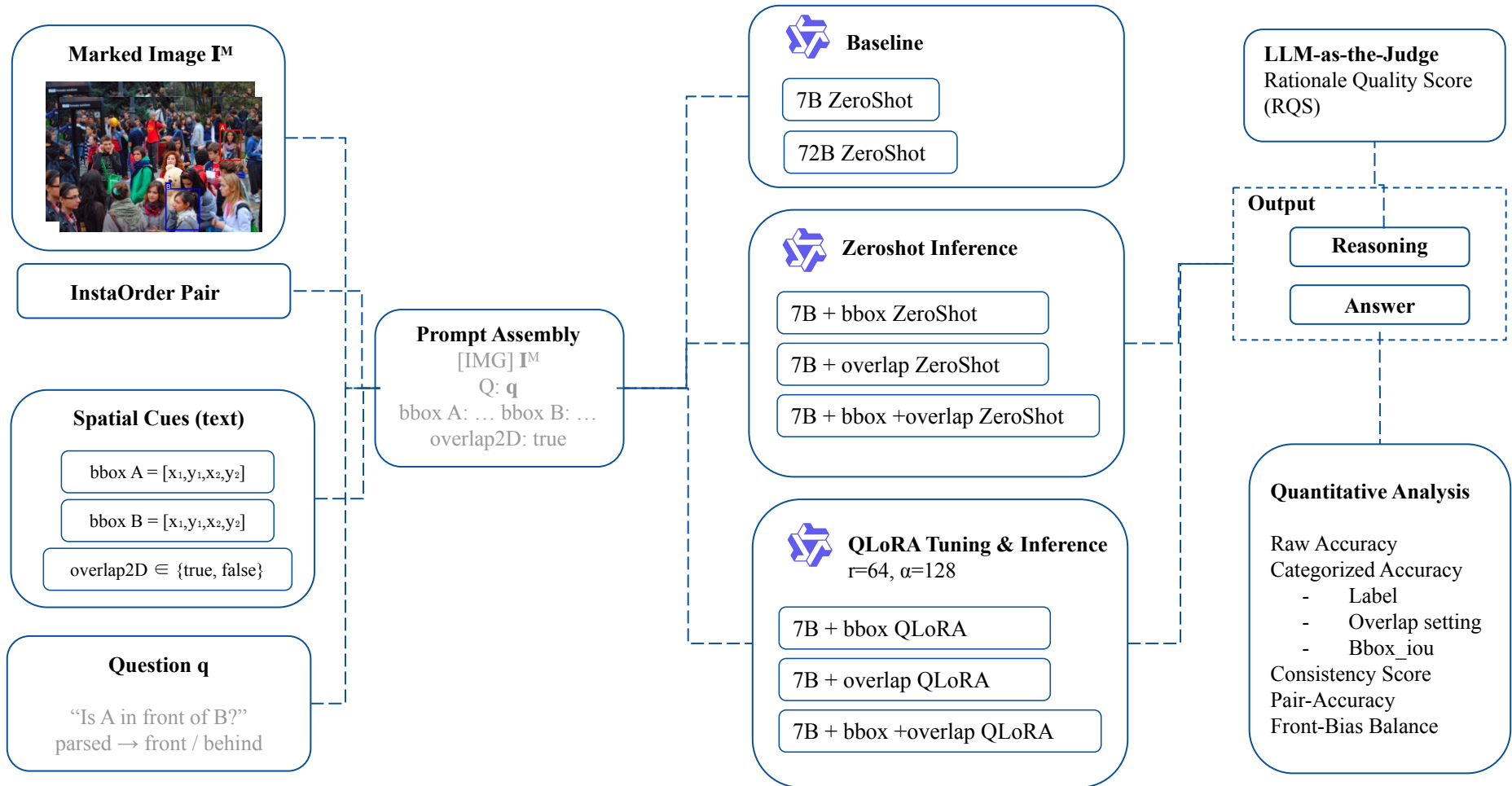


Figure 6, Methodology Workflow

Metric	7B ZS	7B+bbbox	7B+ov	7B+bbbox+ov	72B ZS	7B ZS+bbbox	7B ZS+ov	7B ZS+bbbox+ov
Accuracy								
Raw_Accuracy	50.51%	92.17%	91.57%	92.65%	70.16%	66.26%	65.18%	66.26%
label=front	74.53%	93.16%	93.36%	93.56%	78.52%	80.28%	86.18%	79.20%
label=behind	26.49%	91.19%	89.77%	91.73%	61.79%	52.24%	44.17%	53.32%
overlap=False	53.02%	95.76%	94.90%	96.08%	74.47%	68.72%	67.41%	68.76%
overlap=True	38.20%	74.60%	75.20%	75.80%	49.00%	54.20%	54.20%	54.00%
bbbox_iou								
0 (75.34%)	53.33%	92.72%	92.18%	93.08%	70.64%	66.50%	66.90%	67.57%
(0, 0.1] (14.91%)	45.91%	94.32%	93.18%	93.41%	75.91%	68.51%	63.94%	65.14%
(0.1, 0.3] (7.32%)	38.43%	87.96%	89.81%	91.67%	59.26%	62.50%	53.24%	60.65%
> 0.3 (2.44%)	27.78%	75.00%	68.06%	77.78%	52.78%	56.94%	54.17%	48.61%
Rationale Quality Score (RQS)								
Visual Grounding (1–5)	3.91	4.36	4.81	4.80	4.94	4.61	4.47	4.94
Depth Reasoning (1–5)	3.25	3.84	4.02	4.03	4.51	4.51	4.39	4.61
Hallucination Rate	6.9%	9.6%	5.4%	3.3%	1.8%	0.0%	0.0%	0.1%
Consistency Score	26.15%	95.87%	95.19%	96.82%	67.01%	53.15%	40.81%	50.63%
Pair Accuracy	21.61%	90.11%	89.16%	91.06%	53.66%	45.34%	40.30%	44.33%
Front-Bias Balance	39.94%	98.04%	96.41%	98.17%	83.34%	71.95%	57.95%	74.12%

Table 3, Experiments Result

Key Findings

- **The 7B zero-shot VLM fails in a structured way.** It shows a strong front-label shortcut: relatively high accuracy on front but very poor accuracy on behind.
- **Occlusion and overlap are the central bottlenecks.** Performance drops clearly when objects overlap in 2D, especially in higher bbox_iou buckets.
- **QLoRA with explicit spatial cues substantially improves front-back reasoning.** Adding bbox coordinates and/or overlap flags greatly improves raw accuracy, behind accuracy, pair accuracy, and consistency.
- **The bbox+overlap setting is the strongest overall configuration.** It combines fine-grained localization with coarse pairwise interaction cues, giving the most robust performance.
- **High-overlap cases and reasoning faithfulness remain open challenges.** Future evaluation should test rationale quality and check whether task-specific tuning generalizes to broader VLM benchmarks.

Figure 7, Accuracy By Overlap Setting

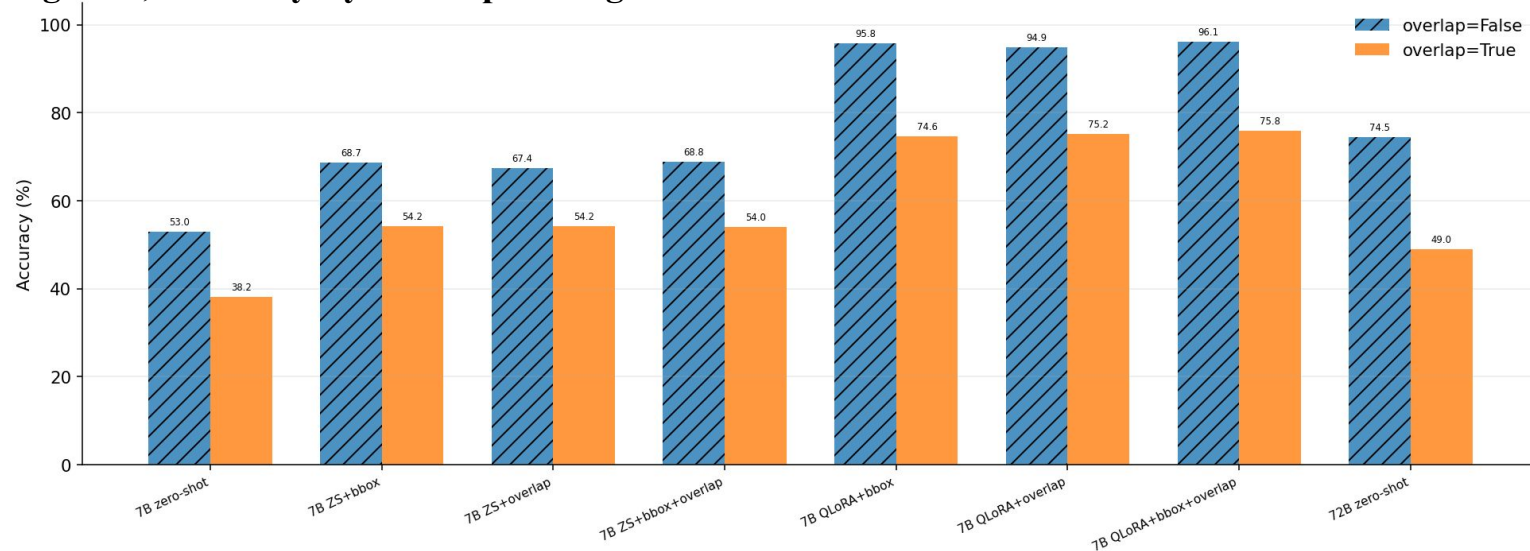


Figure 8, Accuracy By Label

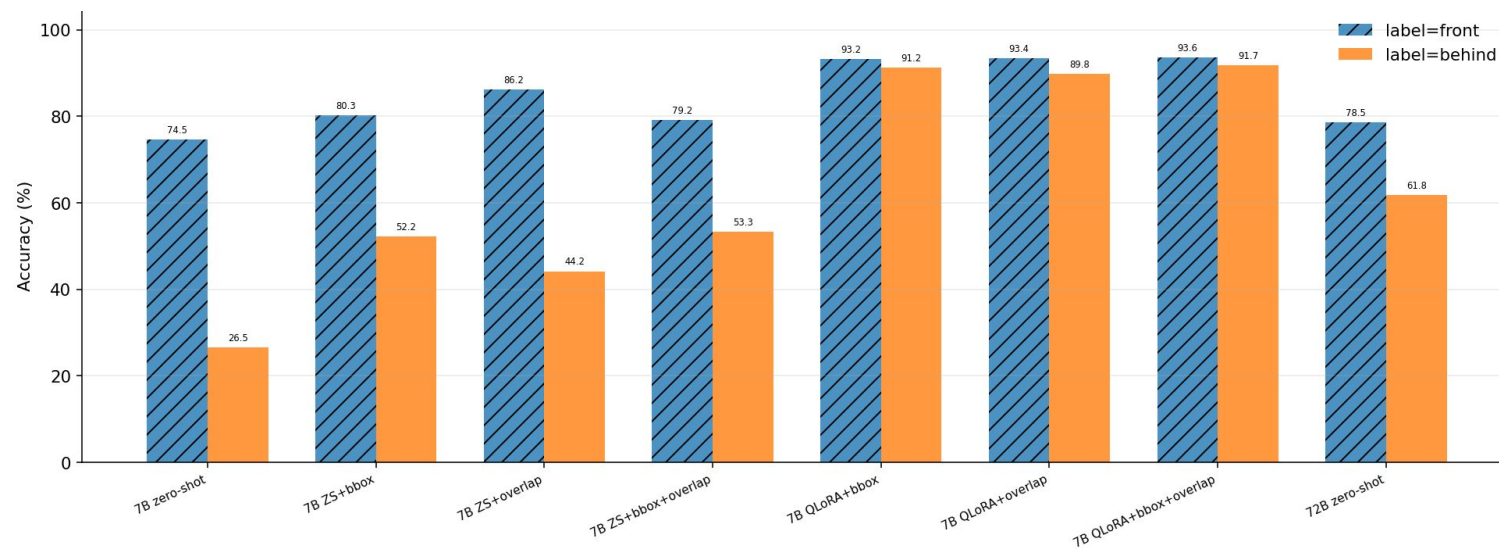


Figure 9, Accuracy By bbox_iou Bucket

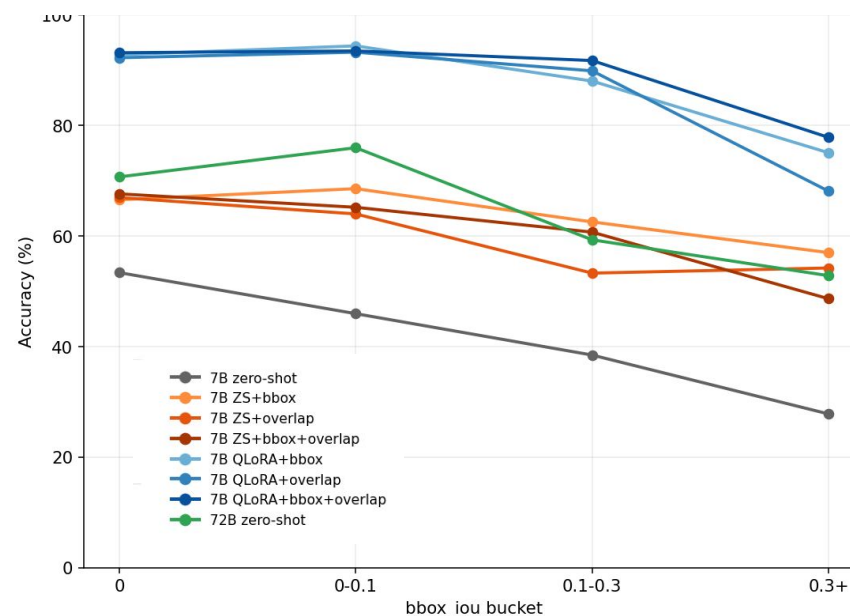


Figure 10, Raw Accuracy vs Consistency Score

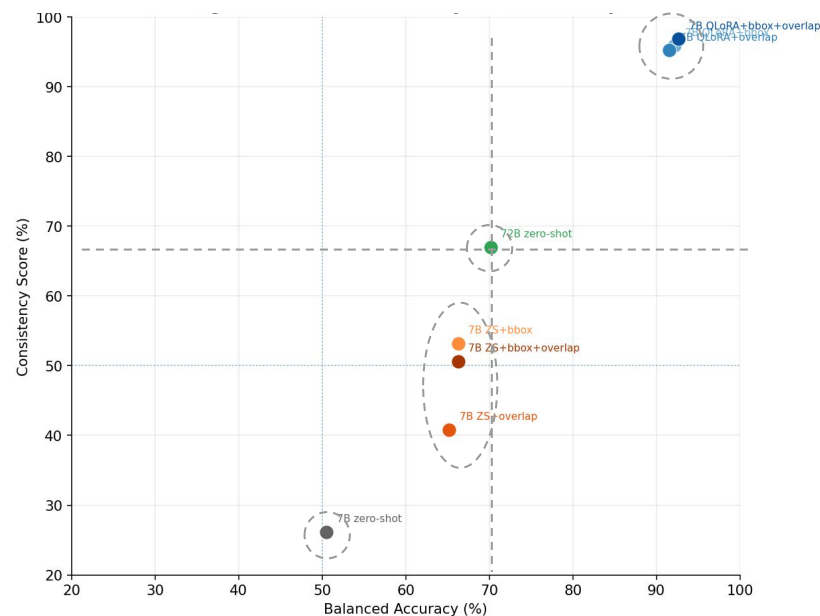
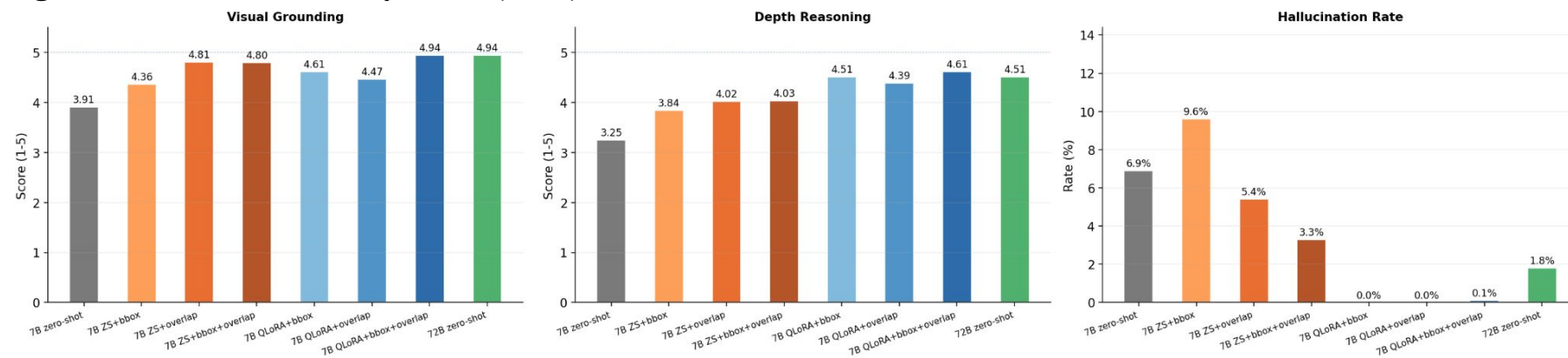


Figure 11, Rationale Quality Score (RQS)



10

Reflection

- Our results show that task-specific spatial cue tuning can substantially improve front-back occlusion reasoning, especially by reducing the 7B model's front-label shortcut.
- However, optimizing a VLM for one narrow spatial task may hurt performance on other benchmark tasks, so broader evaluation is necessary before claiming general spatial improvement.
- The remaining drop in high-overlap cases suggests that lightweight 2D cues help but may still be insufficient for deeply ambiguous occlusion-heavy scenes.

11

Future Work

- Evaluate rationale quality through visual grounding, depth-reasoning validity, and hallucination rate, so we can verify whether higher accuracy reflects genuine spatial reasoning rather than shallow label prediction.
- Test generalization on broader spatial reasoning benchmarks beyond InstaOrder, including non-occlusion relations, embodied spatial tasks, and multi-view reasoning.
- Explore richer visibility-aware cues, such as visible ratio, mask overlap, relative scale, segmentation masks, and depth-order priors.